

A Discrete Dirac–Triality Operator with Octonionic and Hyperbolic Symmetry

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Abstract

We construct a discrete Dirac-type operator combining octonionic triality symmetry, helical translation symmetry, and the discrete group $\mathrm{PSL}(2, 7)$ arising from the Fano plane. The resulting operator exhibits chiral symmetry, band structure, and nontrivial degeneracies induced by exceptional symmetries. Connections to hyperbolic $(2, 3, 7)$ geometry and arithmetic quantum chaos are discussed.

1 Introduction

Exceptional algebraic structures such as the octonions, triality of $\mathrm{Spin}(8)$, and the Fano plane have deep connections with discrete symmetry groups. In particular, the automorphism group of the Fano plane is $\mathrm{PSL}(2, 7)$, which is a quotient of the hyperbolic triangle group $\Delta(2, 3, 7)$.

The aim of this paper is to construct a discrete spectral model combining:

- Helical symmetry $G = \mathbb{Z} \rtimes C_3$,
- Octonionic/Fano symmetry $\mathrm{PSL}(2, 7)$,
- A Dirac-type operator with chiral symmetry.

2 Helical Symmetry Group

Definition 1. *Let*

$$G := \mathbb{Z} \rtimes C_3,$$

where $C_3 = \langle r \mid r^3 = 1 \rangle$ acts on $\mathbb{Z}$ by

$$r \cdot n = n.$$

The group G models discrete screw symmetry: translation along $\mathbb{Z}$ combined with triality rotation.

3 Octonionic Structure and $\mathrm{PSL}(2, 7)$

Let $\mathbb{O}$ denote the octonions with imaginary basis $\{e_1, \dots, e_7\}$ encoded by the Fano plane.

Proposition 1. *The automorphism group of the Fano plane is*

$$\mathrm{PSL}(2, 7),$$

of order 168.

Proof. The Fano plane is the unique projective plane of order 2. Its collineation group is $\mathrm{PGL}(3, 2)$, which is isomorphic to $\mathrm{PSL}(2, 7)$. $\square$

4 Hilbert Space and Representation

We define the Hilbert space

$$\mathcal{H} = \ell^2(G) \otimes \mathbb{C}^4.$$

Here:

- $\ell^2(G)$ carries the regular representation of G ,
- $\mathbb{C}^4$ carries a minimal spinor representation.

5 Dirac–Triality Operator

Let T denote translation on $\ell^2(\mathbb{Z})$:

$$(T\psi)(n) = \psi(n+1).$$

Let R denote the C_3 rotation operator. Let F denote the induced action of $\text{PSL}(2,7)$ on internal degrees of freedom.

Let γ^μ be Dirac matrices satisfying

$$\{\gamma^\mu, \gamma^\nu\} = 2\delta^{\mu\nu}.$$

Definition 2. *The Dirac–Triality operator is*

$$D = \gamma^0 T + \gamma^1 T^{-1} + \gamma^2 R + \gamma^3 F.$$

Proposition 2 (Chiral symmetry). *There exists Γ such that*

$$\Gamma D \Gamma^{-1} = -D.$$

Proof. Choose $\Gamma = \gamma^5$. Standard Clifford algebra relations yield anticommutation with D . $\square$

6 Spectral Analysis

We perform Fourier transform along $\mathbb{Z}$:

$$\psi(n) = \int_{-\pi}^{\pi} e^{ikn} \hat{\psi}(k) dk.$$

Then

$$D(k) = \gamma^0 e^{ik} + \gamma^1 e^{-ik} + \gamma^2 R + \gamma^3 F.$$

6.1 Band Structure

Proposition 3. *The spectrum of D is symmetric about zero.*

Proof. Follows from chiral symmetry. $\square$

For fixed k :

$$D(k)^2 = 2(1 + \cos k)I + (\gamma^2 R + \gamma^3 F)^2 + \text{cross terms}.$$

Eigenvalues satisfy

$$\lambda(k)^2 = 4 \cos^2(k/2) + \mu^2,$$

where μ depends on representation components of R and F .

6.2 Degeneracies

Theorem 1. *Degeneracies in the spectrum of D are governed by irreducible representations of $\mathrm{PSL}(2, 7)$.*

Proof. Since F acts via a representation of $\mathrm{PSL}(2, 7)$, D decomposes according to irreducible components. Multiplicity follows from representation theory. $\square$

6.3 Spectral Statistics

Numerical experiments on finite quotients indicate:

- Nontrivial level clustering,
- Symmetry-induced degeneracies,
- Features reminiscent of arithmetic quantum chaos.

No claim of identification with Riemann zeta zeros is made.

7 Hyperbolic Connection

Since $\mathrm{PSL}(2, 7)$ is a quotient of $\Delta(2, 3, 7)$, the model inherits hyperbolic features. This suggests potential relevance to spectral geometry of arithmetic surfaces.

8 Conclusion

We constructed a discrete Dirac-type operator combining:

- Helical translation symmetry,
- Triality rotation,
- Fano/ $\mathrm{PSL}(2, 7)$ internal symmetry.

The resulting spectral structure provides a mathematically well-defined bridge between exceptional algebra, hyperbolic geometry, and discrete spectral theory.